

CLICK PLUS Ladder Logic — Programming Reference

Powermatic 1200 drill press controls retrofit · CPU C2-01CPU-2 + 2× C2-14D1 · supersedes DWG 05 (Rev H large-format)

Rev I changes (2026-07-20)

1. RUNG 1 corrected: the drum pair (X008 OR X009) is drawn IN SERIES with X011. The Rev H sheet's branch structure read as (X011 OR X008 OR X009) — benign at the motor (ST0/Contactor A were open anyway) but wrong against the §4 definition of C20, and this sheet is the transcription source.
2. Jog rework — hold-FWD is a MODE button, not a jog button:
 - The ≥5 s entry hold arms the mode (green blink) and drives NOTHING. In Rev H the spindle began jogging at the 5 s mark while the entry press was still held; the ~C50 term in RUNG 24 closes that.
 - New RUNG 23A / bit C61 (JOG_RUN_OK): with the drum OFF, mode switching works but the spindle does not follow FWD/REV until the drum is set to LOW or HIGH. Restores "OFF = no motion" inside jog.
3. Boolean expression added under every rung with parallel structure — the ASCII art is illustrative; the Boolean line is authoritative when transcribing branches into CLICK.
4. Changed/new rungs are shaded amber and tagged ◀ REV I.

Symbols

-] [- N.O. -]/[- N.C. -]↑[- press (leading edge) -]↓[- release (trailing edge)
 -()- OUT -(S)- SET -(R)- RESET [TMR name Kvalue] timer ~ = NOT (logic annotations)

Internal bits & timers (Rev I)

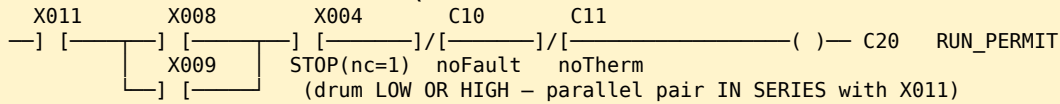
C1 State IDLE (derived: TAP mode AND ~C2 AND ~C3 AND ~fault)
 C2 State TAP_DOWN (latched)
 C3 State BACK_OUT (latched)
 C10 FAULT — latched hard fault (manual clear via RUNG 17)
 C11 THERMAL — DB over-temp, self-clearing (follows X012 with cool-down); inhibits run, blink-red
 C20 RUN_PERMIT = X011 AND (X008 OR X009) AND X004 AND ~C10 AND ~C11
 C30 DRILL_RUN — forward latch
 C31 DRILL_REV — reverse latch
 C40 Chunk-pulse request
 C50 FWD-press-consumed flag — set by mode-entry hold (R19) or FWD-press-stops-REV (R7); while set, the press can neither release-latch a run (R4) nor drive jog motion (R24); clears on release (R8)
 C60 JOG_MODE — mode armed (annunciated: green blink)
 C61 JOG_RUN_OK — jog MOTION permissive (◀ REV I, RUNG 23A): armed AND drum-ON AND STOP-clear AND safe AND no fault/therm. Arming does not require the drum; motion does.
 C99 ~1 Hz free-running blink bit (indicator modulation)
 T1 Chunk pulse timer (300 ms) T5FWD FWD hold timer (5 s → arm jog)
 T5STP STOP hold timer (5 s, jog exit) T60 Jog inactivity timer (60 s → auto-exit)
 Tcool DB cool-down delay (10 s) Tb1/Tb2 Blink astable (500/500 ms → C99)

Scan order is load-bearing — transcribe rungs in the numbered order (23A between 23 and 24).

Rules preserved: (a) DRILL latch-SET rungs (4, 5) run before the opposite-stop rung (7); (b) C50 is set (R7/R19) before the FWD release-latch (R4) and jog legs (R24) evaluate it, and cleared last (R8); (c) C61 (R23A) is computed before the run-output rungs (R24/25) consume it.

PERMISSIVES & HOUSEKEEPING

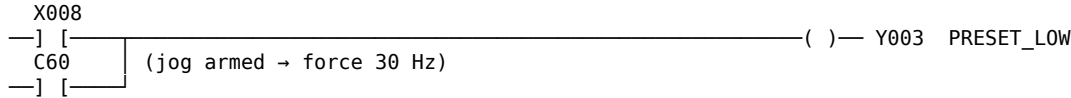
RUNG 1 · RUN PERMISSIVE → C20 (safe AND drum-not-OFF AND STOP-clear AND no fault AND no over-temp)



C20 = X011 AND (X008 OR X009) AND X004 AND NOT C10 AND NOT C11

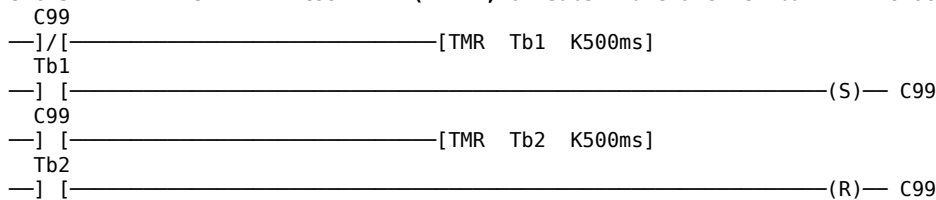
◀ REV I: Rev H sheet drew X011/X008/X009 as a 3-way OR; drum is now correctly in series with safety.

RUNG 2 · SPEED PRESET → Y003 (drum LOW → 30 Hz; jog forces LOW)



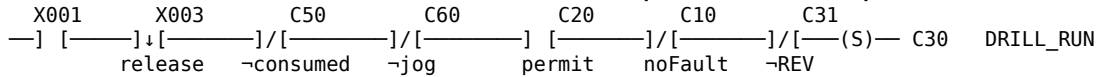
Y003 = X008 OR C60

RUNG 3 · BLINK ASTABLE → C99 (~1 Hz; or substitute a CLICK built-in clock relay and delete this rung)



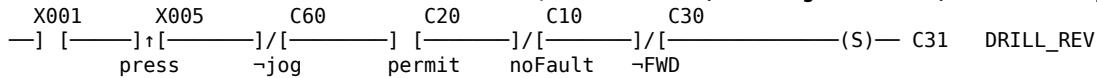
DRILL — LATCHED, BIDIRECTIONAL (stop-first)

RUNG 4 · DRILL FWD — LATCH SET on RELEASE (short tap latches; ~C50 = press not consumed; ~C60 = not in jog)



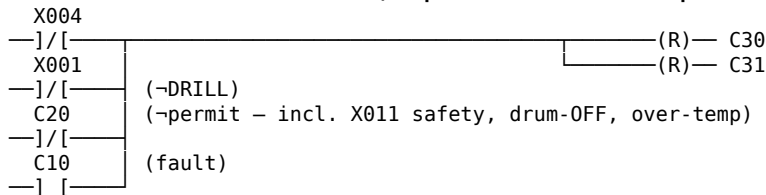
SET C30 when: X001 AND X003↓ AND NOT C50 AND NOT C60 AND C20 AND NOT C10 AND NOT C31

RUNG 5 · DRILL REV — LATCH SET on PRESS (mirror of FWD; no long-hold role, so acts on press)



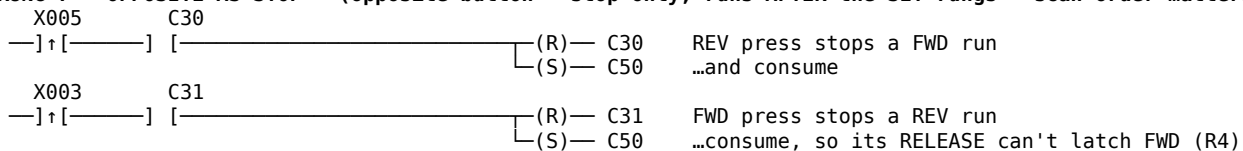
SET C31 when: X001 AND X005↑ AND NOT C60 AND C20 AND NOT C10 AND NOT C30

RUNG 6 · DRILL — LATCH RESET (stop / not DRILL / lost permit / fault → drop both)



RESET C30,C31 when: NOT X004 OR NOT X001 OR NOT C20 OR C10

RUNG 7 · OPPOSITE-AS-STOP (opposite button = stop only; runs AFTER the SET rungs – scan order matters)



(Why it works: R5's ~C30 / R4's ~C31 interlocks block the opposite latch on the SAME press because the running latch is still set when SET evaluates; R7 then drops it. A second press latches.)

RUNG 8 · CLEAR FWD-CONSUMED ON RELEASE

X003
—]/[—————(R)— C50

RUNG 9 · DRILL BOTTOM-LIMIT FAULT (FWD only; reverse is not a feed-down runaway)

—] [——] [——]/[—————(S)— C10 FAULT

SET C10 when: X001 AND C30 AND NOT X006 (X006=0 → at bottom or wire break)

TAP — SINGLE-TAP STATE MACHINE

RUNG 10 · TAP START (IDLE → TAP_DOWN) entry gate: lever OFF (X010=1), not in jog

X002 X003 X010 C60 C20 C10 C2 C3
 — [—] ↑ [—] [—] / [—] [—] / [—] / [—] (S) — C2 TAP_DOWN
 leverOFF ~jog permit noFault

SET C2 when: X002 AND X003↑ AND X010 AND NOT C60 AND C20 AND NOT C10 AND NOT C2 AND NOT C3

RUNG 11 · TAP_DOWN → BACK_OUT (bottom limit)

C2 X006
—] [——]/[————(R)— C2
 (S)— C3 BACK OUT

RUNG 12 · BACK_OUT → IDLE (top limit) [note C: also SET C2 here for continuous tapping]

$$\text{C3} \quad \text{X007}$$
$$-] [-] / [- (R) - \text{C3}$$

RUNG 13 · TAP ABORT (stop / not TAP / fault / lost-permit-or-safety)

X004
 —] / [————— (R) — C2
 X002
 —] / [————— (R) — C3
 (¬TAP)
 C10
 (fault)
 —] [—————
 C20
 (lost permit – incl. E-stop/safety via X011, drum-OFF, STOP)
 —] / [—————

RUNG 14 · TAP FEED-LEVER RUNTIME TRIP (lever ON during a cycle = tap-snap)

C2 X010
 — [—] / [————— (S) — C10 FAULT
 C3 LeverON
 — [—]

DB OVER-TEMP (self-clearing) & FAULT RESET

RUNG 15 · DB OVER-TEMP SET (X012=0 = over-temp OR wire break → assert immediately)

X012
—]/[—————(S)— C11 THERMAL

RUNG 16 · DB COOL-DOWN → SELF-CLEAR (clears only after X012 stays normal for the cool-down delay)

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X012
—] [—————[TMR Tcool K10000ms]
Tcool
—] [—————(R)— C11

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(Self-clearing: no manual reset; C11 is NOT the operator-clearable C10, so it can't fight R17.)

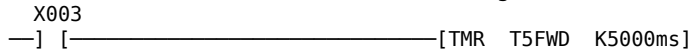
RUNG 17 · FAULT RESET (STOP while fully stopped clears the hard-fault latch)

X004 C2 C3 C30 C31 (R) C10 FAULT

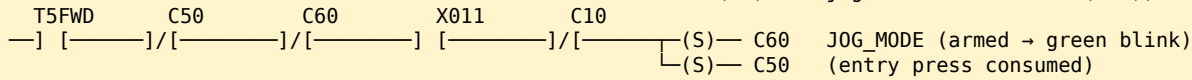
JOG MODE ◀ **REV I:** hold-FWD is a **MODE** button — the entry hold never drives the spindle;

drum OFF permits mode switching but inhibits jog MOTION (see C61, RUNG 23A).

RUNG 18 · FWD HOLD TIMER (times how long FWD is held)

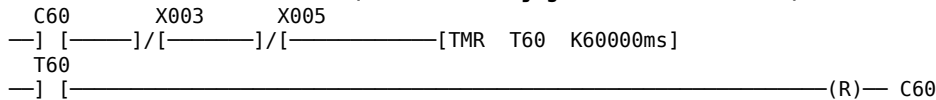


RUNG 19 · ENTER JOG MODE (FWD held ≥ 5 s, safe, not already armed; consume the press so it can neither latch a run on release (R4) nor jog while still held (R24))

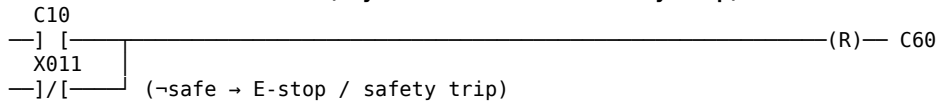


Arming is allowed with the drum OFF – motion is not (C61, R23A). ◀ REV I commentary

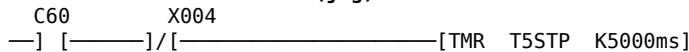
RUNG 20 · JOG IDLE TIMEOUT (60 s with no jog button → auto-exit)



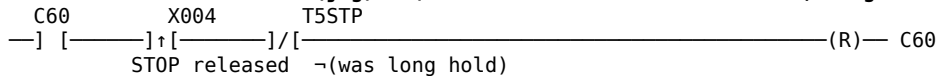
RUNG 21 · JOG FORCE-EXIT (any latched fault or safety trip)



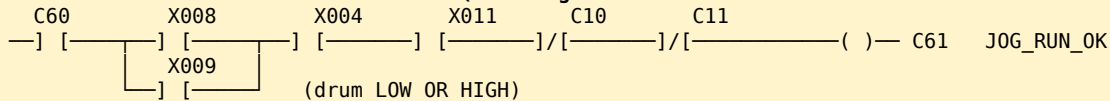
RUNG 22 • STOP HOLD TIMER (jog)



RUNG 23 · SHORT-STOP EXIT (jog) (STOP released before 5 s → exit; long hold = plain stop, stays armed)



RUNG 23A · JOG RUN PERMISSIVE → C61 (motion gate: armed AND drum ON AND STOP clear AND safe AND healthy)

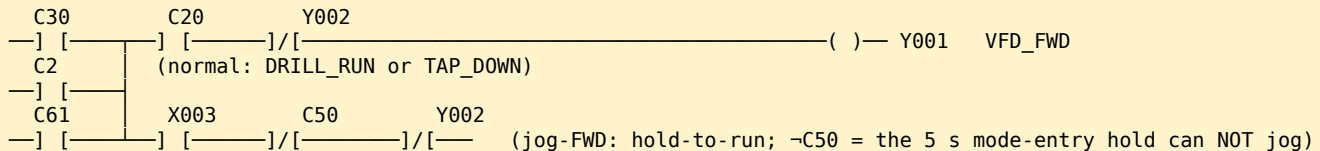


C61 = C60 AND (X008 OR X009) AND X004 AND X011 AND NOT C10 AND NOT C11

▶ **REV I (new rung):** with the drum OFF you can arm/disarm jog, but the spindle will not move until the drum is set to LOW or HIGH. Restores the "OFF = no motion" contract of §3.1 inside jog mode.

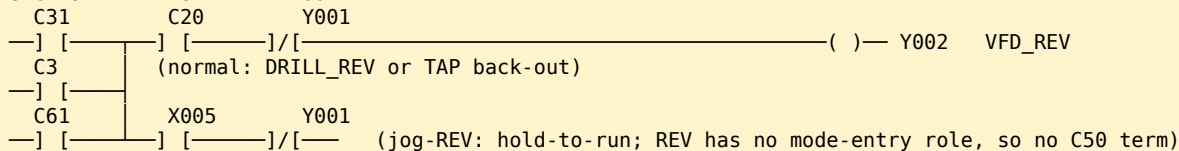
VFD RUN OUTPUTS

RUNG 24 • VFD RUN-FWD → Y001

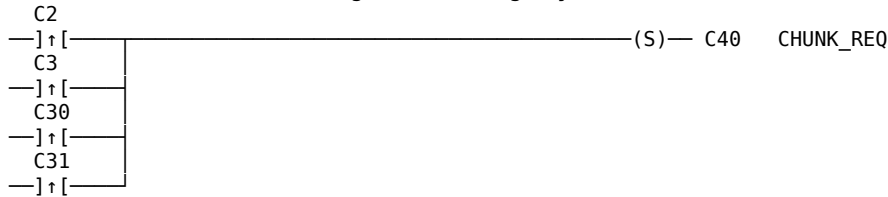
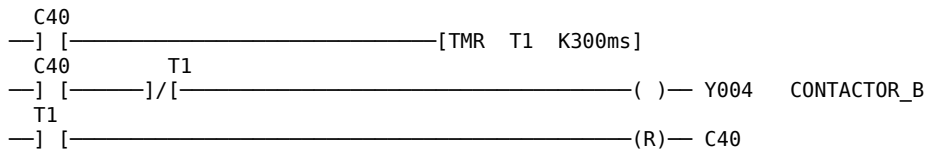
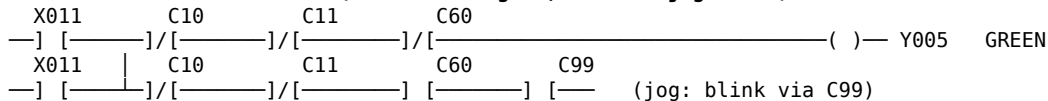

$$Y001 = [(C30 \text{ OR } C2) \text{ AND } C20 \text{ AND NOT } Y002] \text{ OR } [C61 \text{ AND } X003 \text{ AND NOT } C50 \text{ AND NOT } Y002]$$

► **REV I: jog leg gated by C61 (adds drum permissive) and by -C50 – in Rev H the spindle began jogging**
at the 5 s mark while the entry press was still held. The mode-entry hold now drives nothing.

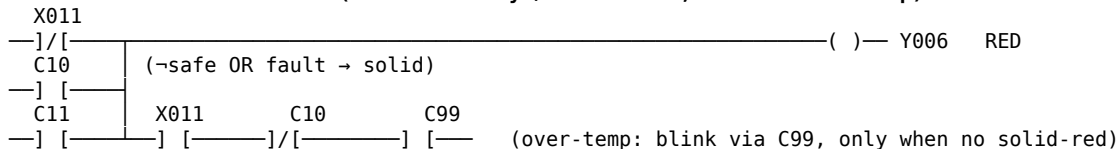
RUNG 25 • VFD RUN-REV → Y002


$$Y002 = [(C31 \text{ OR } C3) \text{ AND } C20 \text{ AND NOT } Y001] \text{ OR } [C61 \text{ AND } X005 \text{ AND NOT } Y001]$$

◀ REV I: jog leg gated by C61 (adds drum permissive).

CHUNK (Contactor B — theater)**RUNG 26 · CHUNK REQUEST (edge of entering any run / direction – TAP or DRILL)****RUNG 27 · CHUNK TIMER + OUTPUT****INDICATORS (4 states, exactly one active)****RUNG 28 · GREEN → Y005 (solid = all-good; blink = jog armed)**

Y005 = [X011 AND NOT C10 AND NOT C11 AND NOT C60] OR [X011 AND NOT C10 AND NOT C11 AND C60 AND C99]

RUNG 29 · RED → Y006 (solid = safety / hard fault; blink = over-temp)

Y006 = [NOT X011] OR [C10] OR [C11 AND X011 AND NOT C10 AND C99]

LADDER NOTES

- A · Force LOW in TAP (optional). R2 is Y003 = X008 OR C60. To also force LOW in every TAP cycle, add OR C2 OR C3. Left off so the drum is honored in TAP.
- B · FWD/REV interlock. R24/R25 each carry the other output's N.C.; the drive ramps through zero on reversal, so no PLC dead-time is required. Direction changes in DRILL are stop-first (R5/R6/R7), so the interlock only guards simultaneous-edge corner cases.
- C · Continuous tapping. In R12 also SET C2 when it RST C3 (top → back down). One rung.
- D · First scan. IDLE is derived and every latch (C2 C3 C10 C11 C30 C31 C60) comes up cleared, so the machine powers up stopped, out of jog, and valid – no init rung.
- E · Max-dwell watchdog (optional). Time C2/C3; if a state overruns its limit, SET C10. Backstops the VFD over-torque trip and E-stop; does not replace them.
- F · C50 is the shared "this FWD press is spent" flag (Rev I: now triple-duty). Set when a 5 s hold arms jog (R19) or when a FWD press stops a running REV (R7). While set, the press can neither latch a forward run on release (R4) nor drive jog motion (R24). Clears on FWD release (R8). Scan order matters: R8 runs AFTER R4 and R24, so the flag is consumed before it clears.
- G · Blink source. R3 is a portable 1 Hz astable; CLICK PLUS also exposes a built-in clock-pulse relay you can assign to C99 and delete R3.
- H · Jog is advisory. Nothing in the jog rungs touches the safety path. An E-stop opens ST0 and drops Contactor A regardless of mode; R21 force-exits jog on the resulting ~X011.
- I · Why exit is via STOP, not a second 5 s FWD hold (Rev I). In jog, holding FWD is hold-to-run – a legitimate jog can exceed 5 s, so a hold-again-to-exit would kick you out of the mode mid-jog. Exit paths: short (<5 s) STOP press · 60 s idle · any latched fault · E-stop/safety trip.

REV I VALIDATION DELTAS (add to §10 checklist)

- [] With safety tripped (X011=0) and drum at LOW or HIGH, C20 stays OFF (verifies R1 series fix).
- [] Hold FWD ≥ 5 s with drum ON: green goes to blink at 5 s and the spindle does NOT turn while the entry press is still held; release, then a fresh FWD press jogs. (Verifies $\neg$ C50 in R24.)
- [] With drum OFF: hold FWD ≥ 5 s $\rightarrow$ jog arms (green blink); FWD/REV presses do nothing; set drum to LOW $\rightarrow$ presses now jog at 30 Hz. Return drum to OFF mid-hold $\rightarrow$ motion stops. (Verifies C61.)
- [] Short STOP press exits jog; long STOP hold stops motion but stays armed; 60 s idle disarms.